

# Yuan Gao

Ph.D. Candidate & Research Associate — Autonomous Vehicle Systems  
Technical University of Munich

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🔗 [yuangao-tum](https://yuangao-tum.github.io) | 🎓 [Google Scholar](https://scholar.google.com/citations?user=yuan-gao-tum) | [in](https://www.linkedin.com/in/yuan-gao-tum) [yuan-gao-tum](https://www.linkedin.com/in/yuan-gao-tum) | 📄 [ResearchGate](https://www.researchgate.net/profile/Yuan-Gao)



🎯 **Actively seeking a Ph.D. research internship** in foundation models for autonomous driving — foundation-model-based scenario generation & analysis, and vision-language-action (VLA) models.  
**Available from early 2027 for 3–6 months.**

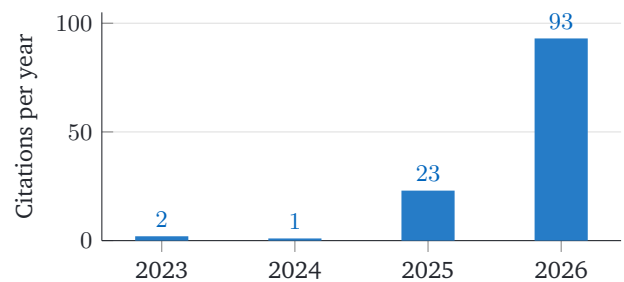
## RESEARCH INTERESTS

Foundation models (LLMs, VLMs, VLAs, world models) for autonomous driving — scenario generation and scenario analysis, safety-critical scenario evaluation, visual question answering and benchmark construction, semantic anomaly detection, agent-level risk assessment; motion planning and optimal control of multi-agent systems.

## RESEARCH IMPACT

- 📄 **121 citations total** | h-index 4 | i10-index 2
- 🔔 **93 citations in 2026 alone** — already ~4× the 2025 total
- ★ **228 GitHub stars** on the FM-AD-Survey repository
- 📄 **16 research outputs** (12 peer-reviewed) — 10 first-authored

Source: [Google Scholar](https://scholar.google.com/citations?user=yuan-gao-tum), August 2026.



2026 is year-to-date; Scholar per-year counts may not sum to the total.

## EDUCATION

### Ph.D. in Mechanical Engineering (Autonomous Vehicle Systems)

Technical University of Munich (TUM), School of Engineering and Design

Professorship of Autonomous Vehicle Systems; advisor: Prof. Dr.-Ing. Johannes Betz.

Research: scenario generation and analysis for autonomous driving based on foundation models.

Oct 2024 – present

### M.Sc. Development, Production, and Management in Mechanical Engineering

Technical University of Munich (TUM)

Grade 1.5 — passed with distinction (double Master's program).

Oct 2021 – Aug 2023

### M.Sc. Mechatronics and Robotics

Technical University of Munich (TUM)

Grade 1.2 — passed with high distinction.

Master's thesis at the Chair of Information-oriented Control (Prof. Dr.-Ing. Sandra Hirche, Dr. Junjie Jiao).

Oct 2019 – Feb 2023

### M.Sc. studies, Mechatronics and Information Technology

Karlsruhe Institute of Technology (KIT)

Continued Master's studies at TUM from Oct 2019.

Oct 2018 – Sep 2019

### B.Eng. Mechanical Design, Manufacture, and Automation

Hefei University of Technology, China

GPA 3.74/4.3, rank 3/384 — Excellent Graduate honors.

Exchange semester at I-Shou University, Taiwan (Feb – Jun 2015), GPA 4.0/4.0.

Sep 2013 – Jun 2017

## AWARDS & HONORS

- Double M.Sc. passed with **high distinction** (Grade 1.2, Mechatronics & Robotics) and **distinction** (Grade 1.5), TUM.
- **Excellent Graduate** honors, Hefei University of Technology — undergraduate rank **3 / 384**.
- Perfect GPA (4.0 / 4.0) on exchange semester, I-Shou University, Taiwan.

## RESEARCH & PROFESSIONAL EXPERIENCE

### Research Associate / Ph.D. Candidate

Oct 2024 – present

*Professorship of Autonomous Vehicle Systems, TUM*

- Lead research on foundation-model-based scenario generation and analysis for autonomous driving in a project funded by the German Central Innovation Programme for SMEs (ZIM).
- First-authored publications at EMNLP, ICRA, IROS, ITSC, and IEEE OJ-ITS; co-authored papers at ECCV (2×) and ITSC; built open-source frameworks, benchmarks, and datasets (PlannerForge, NuRisk, FM-AD-Survey, From Words to Collisions).
- Supervised 13 student theses/projects; teaching assistant for the F1TENTH practical course and the Future Mobility Camp seminar.

### Research Assistant

Jul 2023 – Nov 2023

*Chair of Information-oriented Control, TUM*

Distributed optimal control of multi-agent systems. Afterwards (Dec 2023 – Sep 2024), prepared Ph.D. applications and extended the Master's thesis results into a journal submission (IEEE TCNS, under review).

### Intern — Healthcare / Medical Robotics

Feb 2023 – Jun 2023

*ITK Engineering GmbH (Bosch Group)*

Internship in the healthcare and medical robotics product area.

### Student Research Assistant

Mar 2022 – Aug 2022

*Chair of Automotive Technology, TUM (Prof. Dr.-Ing. Markus Lienkamp)*

Driving-simulator research for automated driving; previously completed a semester thesis at the same chair (May – Dec 2021).

## PUBLICATIONS

Author name in bold; citation counts from Google Scholar, August 2026. 10 of 16 publications are first-authored.

### Journal Articles

- [J1] **Y. Gao**, M. Piccinini, Y. Zhang, D. Wang, K. Moller, R. Brusnicki, B. Zarrouki, A. Gambi, J. F. Totz, K. Storms, S. Peters, A. Stocco, B. Alrifaae, M. Pavone, J. Betz. “Foundation Models in Autonomous Driving: A Survey on Scenario Generation and Scenario Analysis.” *IEEE Open Journal of Intelligent Transportation Systems*, 2026. **82 citations** [DOI] [arXiv] [code ★228]
- [J2] **Y. Gao**, J. Jiao, Z. Li, S. Hirche. “H<sub>2</sub> Suboptimal Containment Control of Homogeneous and Heterogeneous Multi-Agent Systems.” *IEEE Transactions on Control of Network Systems* (under review). [arXiv]

### Conference & Workshop Papers

- [C1] **Y. Gao**, S. Müller, M. Piccinini, M. Kaufeld, Y. Zhang, F. R. Schäfer, Q. Song, J. Betz. “PlannerForge: LLM Agents for Scenario-Based Testing of Motion Planners in Autonomous Driving.” *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026 (main conference, accepted). [arXiv] [code] [project] [dataset]
- [C2] **Y. Gao**, M. Piccinini, R. Brusnicki, Y. Zhang, J. Betz. “NuRisk: A Visual Question Answering Dataset for Agent-Level Risk Assessment in Autonomous Driving.” *IEEE International Conference on Robotics and Automation (ICRA)*, 2026. **6 citations** [arXiv] [code ★7] [dataset]
- [C3] **Y. Gao\***, W. Miao\*, M. Piccinini, H. Wang, Q. Song, J. Betz (\* equal contribution). “Chat2Scenic: An Iterative RAG-Based Framework for Scenario Generation in Autonomous Driving.” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026 (accepted).
- [C4] F. R. Schäfer, **Y. Gao**, D. Wang, T. Stauner, S. Günemann, M. Piccinini, S. Schmidt, J. Betz. “EgoDyn-Bench: Evaluating Ego-Motion Understanding in Vision-Centric Foundation Models for Autonomous Driving.” *European Conference on Computer Vision (ECCV)*, 2026 (accepted). [arXiv]
- [C5] D. Wang, Z. Liang, H. Ye, Z. Sun, Z. Lu, Y. Zhang, Y. Zhao, **Y. Gao**, M. Seegert, F. R. Schäfer, H. Qin, W. Li, L. Palmieri, F. Jahncke, M. Piccinini, J. Betz. “Target-Bench: Can Video World Models Achieve Mapless Path Planning with Semantic Targets?” *European Conference on Computer Vision (ECCV)*, 2026 (accepted). **6 citations** [arXiv] [code] [project] [dataset]
- [C6] R. Brusnicki, D. Pop, **Y. Gao**, M. Piccinini, J. Betz. “Can VLMs Unlock Semantic Anomaly Detection? A Framework for Structured Reasoning.” *IEEE International Conference on Intelligent Transportation Systems (ITSC)*, 2026 (accepted). [arXiv] [code] [project]
- [C7] Q. Song, **Y. Gao**, R. Brusnicki, F. Sarro. “Warnless at the DeepTest 2026 Tool Competition.” *7th IEEE/ACM International Workshop on Deep Learning for Testing (DeepTest)*, 2026. [ACM DL]
- [C8] **Y. Gao**, M. Piccinini, J. Betz. “Towards Whole-Body VLA: A Scalable Data Collection Framework for Quadrupedal Mobile Manipulators.” *2nd German Robotics Conference*, 2026. [PDF]

- [C9] F. R. Schäfer, K. Moller, **Y. Gao**, C. Oefinger, S. Schmidt, J. Betz. “Imagined Rollouts are Kinematic, Not Dynamic: A Diagnosis of Long-Horizon World-Model Failure.” *RSS Robot World Model Workshop*, 2026. [\[arXiv\]](#)
- [C10] **Y. Gao**, M. Piccinini, K. Moller, A. Alanwar, J. Betz. “From Words to Collisions: LLM-Guided Evaluation and Adversarial Generation of Safety-Critical Driving Scenarios.” *IEEE International Conference on Intelligent Transportation Systems (ITSC)*, 2025. [🔗 17 citations](#) [\[arXiv\]](#) [\[code\]](#) [★ 31](#)
- [C11] **Y. Gao**, J. Jiao, S. Hirche. “H<sub>2</sub> Suboptimal Leader-Follower Consensus Control of Multi-Agent Systems.” *IFAC World Congress*, 2023. *IFAC-PapersOnLine* 56(2), 2614–2619. [🔗 3 citations](#) [\[code\]](#)
- [C12] **Y. Gao**, J. Jiao, S. Hirche. “H<sub>2</sub> Suboptimal Containment Control of Multi-Agent Systems.” *American Control Conference (ACC)*, 2023. [\[IEEE Xplore\]](#) [\[code\]](#)

## Preprints

- [P1] Q. Song, **Y. Gao**, J. Betz, D. Pfahl, M. R. Mousavi, F. Sarro. “In the Driver’s Seat: A Multi-Company Study on the Reality of Autonomous Driving System Testing.” *arXiv:2607.15820*, 2026. [\[arXiv\]](#)
- [P2] Z. Li, F. Zhao, M. Piccinini, B. Zarrouki, **Y. Gao**, Z. Shan, J. Betz, C. Lv, L. Xie. “SGTP: Sampling-based Game-Theoretic Planning for Real-Time Multi-Vehicle Autonomous Racing.” *arXiv:2607.25388*, 2026. [\[arXiv\]](#) [\[code\]](#) [\[project\]](#)
- [P3] **Y. Gao**, D. Hua, M. Piccinini, F. R. Schäfer, K. Moller, L. Li, J. Betz. “StyleVLA: Driving Style-Aware Vision Language Action Model for Autonomous Driving.” *arXiv:2603.09482*, 2026. [🔗 3 citations](#) [\[arXiv\]](#)

## TALKS & PRESENTATIONS

- IEEE ICRA 2026** (Vienna, Austria) — presented *NuRisk* [C1]. Jun 2026
- IEEE ITSC 2025** (Gold Coast, Australia) — presented *From Words to Collisions* [C8]; workshop talks at the *LLVM-AD* and *Interaction-driven Behavior Prediction & Planning* workshops. Nov 2025
- Research visit**, Australian Centre for Robotics, University of Sydney. Nov 2025

## ACADEMIC SERVICE

- Conference reviewer:** NeurIPS 2026 • ICLR 2026 • CVPR 2026 • ICRA 2026, 2025 • IROS 2026 • ITSC 2026, 2025 • IEEE IV 2026, 2025 • IFAC World Congress 2023.
- Journal reviewer:** IEEE Open Journal of Intelligent Transportation Systems • IEEE Robotics and Automation Letters.

## TEACHING (TECHNICAL UNIVERSITY OF MUNICH)

- Teaching Assistant**, “F1TENTH: Autonomous Driving Hands-on” (practical course, English). WS 2024, WS 2025
- Teaching Assistant**, “Future Mobility Camp” (project-week seminar, practical sessions). WS 2024, WS 2025
- Tutor**, “F1TENTH: Autonomous Driving Hands-on” (practical course, English). WS 2024
- Tutor**, “The Driving Simulator in the Development Process” (practical course, German). SS 2022
- Certification:** TUM Onboarding Program, elective module “Designing Courses.” Dec 2024

## STUDENT SUPERVISION (13 THESES & PROJECTS)

- Yang Zhang, *Ethics-Aware Decision Modeling for Autonomous Driving Using Large Vision-Language Models*, Master’s thesis, Feb – Aug 2026
- Yuheng Liu, *CoVLM: A Vision-Language-Model-Based Semantic V2V Cooperation Agent for Autonomous Driving*, Master’s thesis, Feb – Aug 2026
- Chen Li, *Multimodal Retrieval-Augmented Closed-Loop Autonomous Driving Scenario Generation*, Master’s thesis, Jun – Dec 2025 – Jun 2026
- Joshua Pusch, *A VLM-Based Scenario Analysis of Autonomous Driving*, interdisciplinary project, Mar – Sep 2025 – Mar 2026
- Haoyu Wang, *Closing the Loop: Integrating Vision-Language Models for Feedback-Driven ADAS Scenario Generation*, Master’s thesis, Mar – Sep 2025 – Mar 2026
- Mengchen Cai, *From State Machines to Traffic Safety: Benchmarking Driver Assistance Systems Using Structured Scenario Models*, research internship, Aug 2025 – Feb 2026
- Karmishth Annauth, *Literature Study: Lightweight and Fast-Inference Vision-Language Models for Autonomous Driving*, semester thesis, Aug 2025 – Feb 2026
- Wenting Miao, *Retrieval-Augmented Generation of Simulator-Compatible ADAS Scenarios*, Master’s thesis, Jul 2025 – Jan 2026
- Dengyuan Hua, *VLM-based Trajectory Generation: Driving Behavior Diversity*, Master’s thesis, Jul 2025 – Jan 2026
- Yung Loon Lim, *Bridging ADAS/ADS Functionality, Regulations, and Scenario-Based Testing: A Comprehensive Literature Review*, semester thesis, Jun – Dec 2025
- Leonardo Zambrano, *Fine-Tuning Vision-Language Models for Risk Assessment in Autonomous Driving*, Bachelor’s thesis, Apr – Oct 2025
- Sebastian Müller, *Generation of Safety-Critical Scenarios Based on LLMs*, Bachelor’s thesis, Mar – Sep 2025
- Marek Lorenz, *Risk-Aware Scenario Analysis Based on LLMs in Autonomous Driving*, Bachelor’s thesis, Oct 2024 – Apr 2025

OPEN-SOURCE SOFTWARE

- [FM-AD-Survey](#) (★ 228) — curated paper collection on foundation models for scenario generation & analysis in autonomous driving.
- [From-Words-to-Collisions](#) (★ 31) — LLM-guided evaluation and adversarial generation of safety-critical scenarios (ITSC 2025).
- [IFAC2023](#) (★ 24) & [ACC2023](#) — MATLAB implementations of H<sub>2</sub> suboptimal multi-agent control papers.
- [Riskaware-Scenario-analyse](#) (★ 21) — risk-aware scenario analysis based on LLMs.

SKILLS

|                        |                                                                                    |
|------------------------|------------------------------------------------------------------------------------|
| Programming            | Python, C, C++, C#                                                                 |
| ML / Foundation Models | PyTorch, Hugging Face, LLMs & VLMs, agentic AI (LangChain, LangGraph, RAG)         |
| Robotics & Systems     | ROS/ROS 2, Linux, Git, CARLA simulation                                            |
| Modeling & CAD         | MATLAB/Simulink, Modelica, SILAB, PLC (CODESYS); AutoCAD, CATIA, Creo, Solid-Works |
| Languages              | Chinese (native) • German (fluent, C1/DSH-2) • English (fluent, C1/IELTS 7.0)      |